

2023 AMC 10B Problem 12 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 12. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 12

**Problem:**

When the roots of the polynomial

$$P(x) = (x - 1)^1(x - 2)^2(x - 3)^3 \cdots (x - 10)^{10}$$

are removed from the real number line, what remains is the union of 11 disjoint open intervals. On how many of these intervals is $P(x)$ positive?

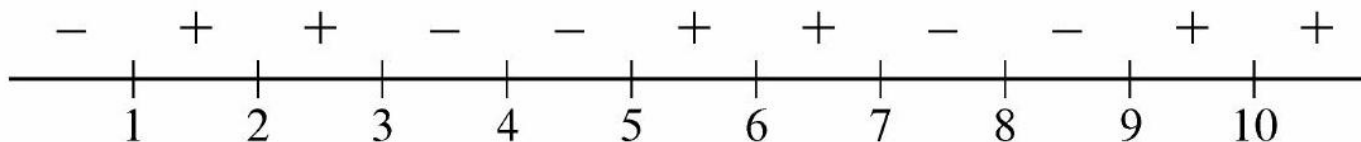
Answer Choices:

- A. 3
- B. 7
- C. 6
- D. 4
- E. 5

Solution:

Note that the function $P(x) > 0$ for $x > 10$. Moving right to left along the real number line, $P(x)$ changes sign precisely when passing through a root of odd multiplicity. This means that the function changes sign as x passes through 9, 7, 5, 3, and 1, and its value is 0 at 10, 8, 6, 4, and 2 as well.

Therefore $P(x) > 0$ on the intervals $(10, \infty)$, $(9, 10)$, $(6, 7)$, $(5, 6)$, $(2, 3)$, and $(1, 2)$, a total of (C) 6 intervals.



OR

Perfect square factors are always nonnegative, so they can be removed without changing the signs. Therefore, for any x with $P(x) \neq 0$, the polynomial

$$Q(x) = (x - 1)(x - 3)(x - 5)(x - 7)(x - 9)$$

will have the same sign as $P(x)$. Now $Q(x)$ is positive on the following intervals: $(1, 2)$ and $(2, 3)$ (where four of the factors are negative), $(5, 6)$ and $(6, 7)$ (where two of the factors are negative), and $(9, 10)$ and $(10, \infty)$ (where all the factors are positive). This gives a total of (C) 6 intervals on which $Q(x) > 0$. On the other 5 intervals there are an odd number of negative factors, so $Q(x)$, and therefore $P(x)$ as well, is negative.

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